Ethics of AI in software development and location-based systems

Petteri Mäkelä

SEAMK

Seinäjoki University of Applied Sciences



11/4/2026

Petteri Mäkelä
Principal lecturer

SEAMK

Introduction

- Petteri Mäkelä
- Principal lecturer, Computer Science
  - Seinäjoki University of Applied Sciences
  - petteri.makela@seamk.fi
- Teaching
  - Programming (Data structures and algorithms, web-based systems, …)
- Teaching and research
  - Industrial Internet of Things
  - Positioning systems (GNSS, indoor positioning)
  - Autonomous mobile robots


Norwegian Sea
Iceland
Sweden
Finland
Norway
Denmark
United Kingdom
Ireland
Belarus
Poland
Germany
Ukraine
Austria
France
Romania
Italy
Spain
Greece
Türkiye
Portugal

Finland

- Population 5.5 million
- 18 inhabitants / 2 km^2
- 7th largest country in Europe
- 187 888 lakes
- Official languages: Finnish (88.9%) and Swedish (5,3%)

- Independent republic since D ecember 6th, 1917
- Member of the European Union (EU) since 1995
- Member of the North Atlantic Treat Organization (NATO) since 2023







City of Seinäjoki

- +66 000 inhabitants.
- Rapidly growing city area.
- The region is well-known for its cultural and entrepreneurial atmosphere.
- One of the most rapidly growing urban areas in Finland.
- Distance between Seinäjoki and Helsinki:
  - 357 km; by train 2 h 35 min, by car 4 hours





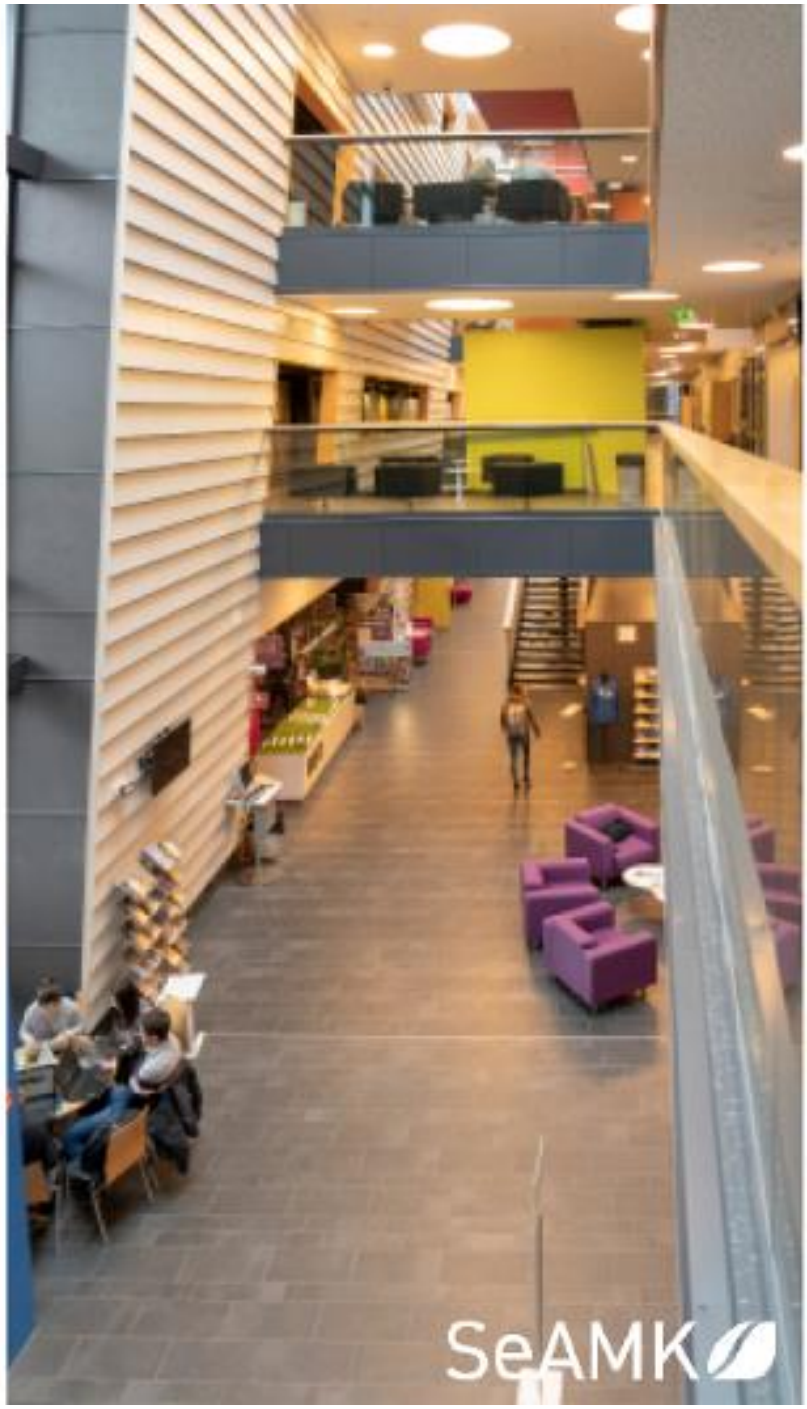














SeAMK Campus

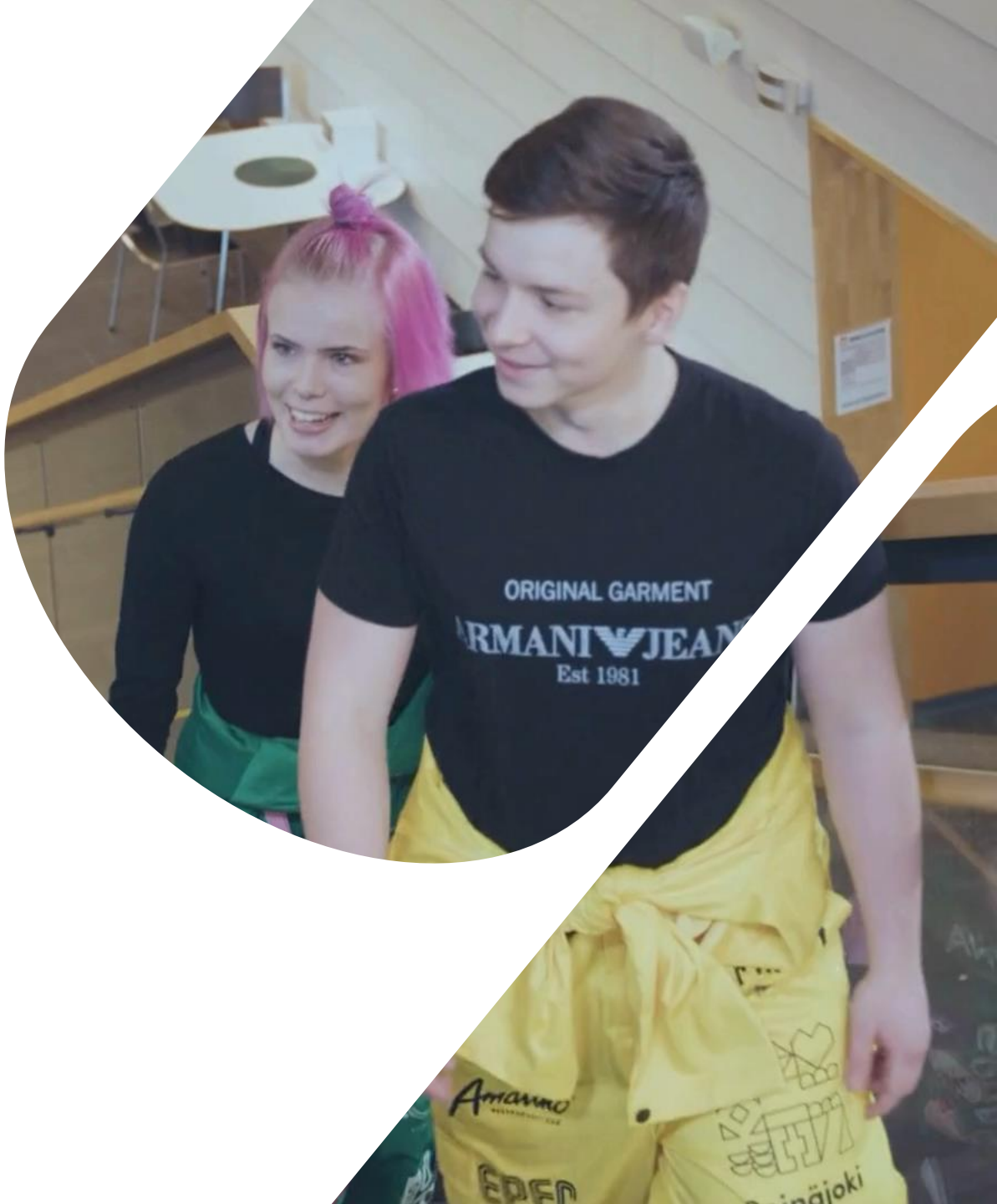


SEAMK facts

- approx. 5 000 students
- +800 graduating students annually
- 250 international mobility students
- 500 international degree students
- +400 employees

Faculties

- Business
- Culture
- Food and Hospitality
- Health Care and Social Work
- Agriculture
- Engineering



SEAMK

AI in software development

From code completion to coding agents

Contents

- How software development has evolved from AI-assisted coding to agentic software development
- Workshop: Build a location-based application with an AI (agent)
  - You act as the product owner. The agent writes the code
- Discussion: ethics of AI in software development
- Location-based systems
- Discussion: ethics of location-based systems

AI in software development today

- Almost all developers use or plan to use AI tools (84 %, Stack Overflow Developer Survey 2025).
- Trust lags behind use. More developers distrust the accuracy of AI output (46 %) than trust it (33 %).
- The big change in 2025–2026:
  - 2025 and before: AI made code suggestions and helped to code faster
  - Now AI agents do the work and people define the goal and check the result
- Developers still need to understand code, architecture and the problem being solved. Now the developers guide and review the output of the agent.

From code completion to coding agents

Last year's workshop ended here

2021 →

Autocomplete

Suggests the next lines while you type

Developer writes, AI finishes

2023 →

Chat assistant

Answers questions and generates code snippets

Developer copies and integrates

2025 →

Coding agent

Reads the codebase, plans, edits many files, runs tests and fixes its own errors

Developer sets the goal and reviews

Late 2025 →

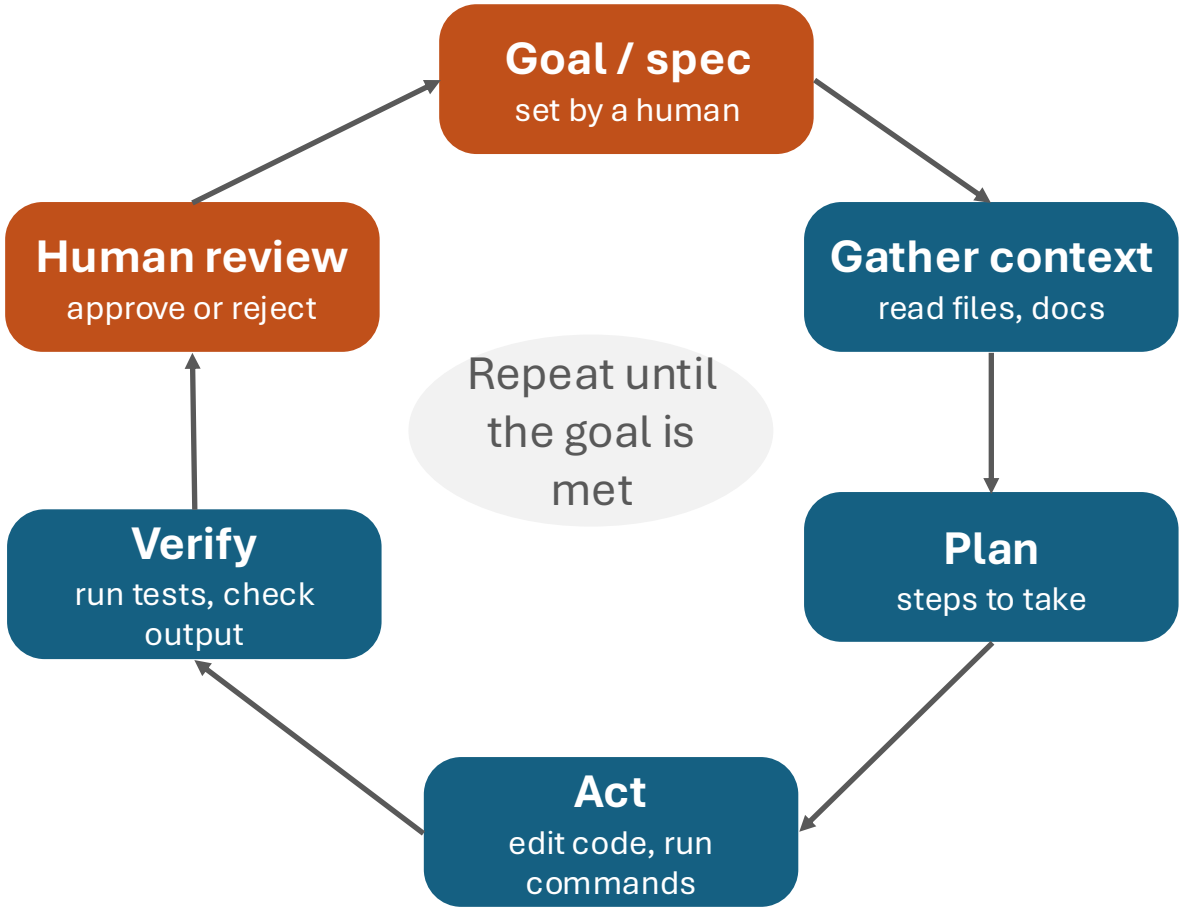
Background agents

Work on tickets in the cloud, in parallel, and return pull requests

Team manages a fleet of agents

- AI no longer only produces text. It takes actions, such as, runs the programs, reads the results, tries again, etc.)

How a coding agent works
The agent loop



An agent is a language model running in the loop with tools. It keeps working until the goal is met or it gets stuck.

Tools : The agent acts through tools: files, terminal, browser, version control systems. Agent is also connected to tickets, databases and other systems.

Context files: AGENTS.md tell the agent the project rules: code style, how to run and test, what not to touch.

Guardrails: Permissions, sandboxes and limits on what the agent may access or delete.

Verification: Agent runs the program and tests it. An agent is only as good as its feedback. Automated tests matter more than before.

■ human ■ agent

What agents do and what people do

Coding agents can

- Read and understand a whole codebase
- Plan a change and edit many files
- Run programs, tests and Git commands
- Find and fix their own errors
- Write tests and documentation
- Work on a ticket in the background and open a pull request

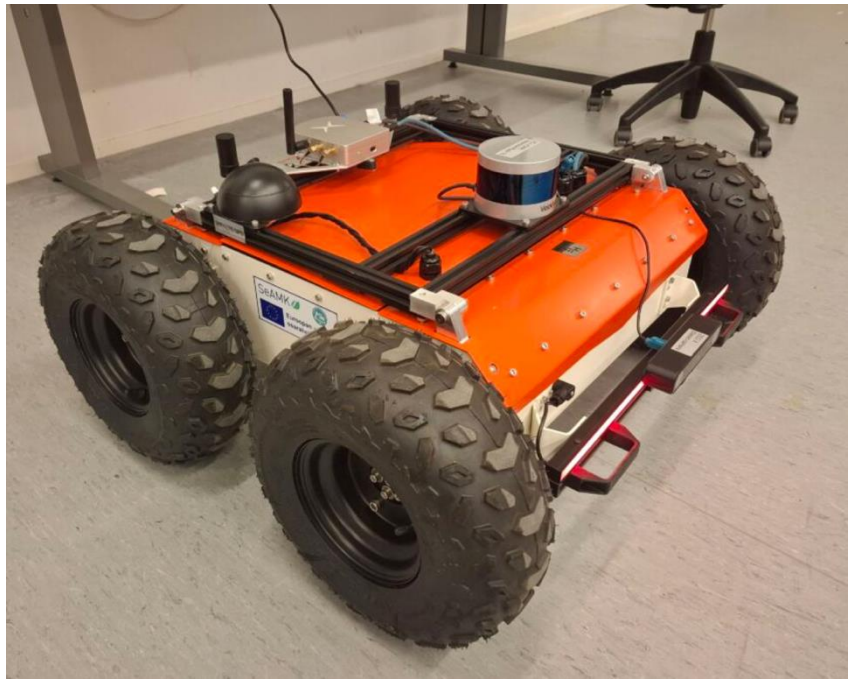
People are still needed to

- Decide what to build and why
- Write clear specifications and acceptance criteria
- Review, test and approve changes
- Own architecture, security and data protection
- Take responsibility for the result

According to Anthropic (Claude)

What people do

- The previous slide was Anthropic’s view
- People have still to find and solve the problems when agents can’t do that
- Automatic testing and problem solving is not always possible in embedded systems (e.g., control of machines)
- People have still to have wide knowledge computer science and IT systems:
  - Programming languages, algorithms, architecture patterns, networks, data communication, servers, cloud services, sensors, devices, etc.



Husarion Panther AMR

Vibe coding vs. agentic engineering (by Anthropic)

Vibe coding

“Forget that the code even exists” (Karpathy, 2025)

- Describe what you want, accept what the AI produces
- Nobody reads the code
- Testing = “does it seem to work?”
- Fast and fun for prototypes and personal tools
- Hard to maintain, risky in production

Agentic engineering

AI does the work, people stay in control

- Start with a specification and acceptance criteria
- The agent proposes a plan before coding
- Small steps, automated tests, version control
- A person reviews and approves every change
- Suitable for production software

New risks with coding agents

- Security: Agents get wide access to files, terminal and internet. Malicious instructions hidden in web pages, issues or documents (prompt injection) can mislead them. Secrets may leak.
- Quality: Large amounts of code that nobody fully understands. Code review becomes the bottleneck. AI may generate incorrect or inefficient code. Code reuse may decrease.
- Accountability: Who is responsible when agent-written code fails. The developer, the company or the tool provider?
- Cost and energy: One agent task can use thousands of times more computing than a single chat answer. Using agentic AI may become expensive.

New roles in agentic software development

- Developer
  - From writer of code to specifier, reviewer and orchestrator of agents
- Product owner and business
  - Can build working prototypes themselves
  - The specification becomes the most important input for SW development
- Team and organization
  - Review and testing become the bottleneck.
  - Decide which tools, data and permissions agents get.

Discussion

If agents write the code, what should a business student and a computer science student each learn? Is the developer role described above realistic?

Productivity and AI software engineering

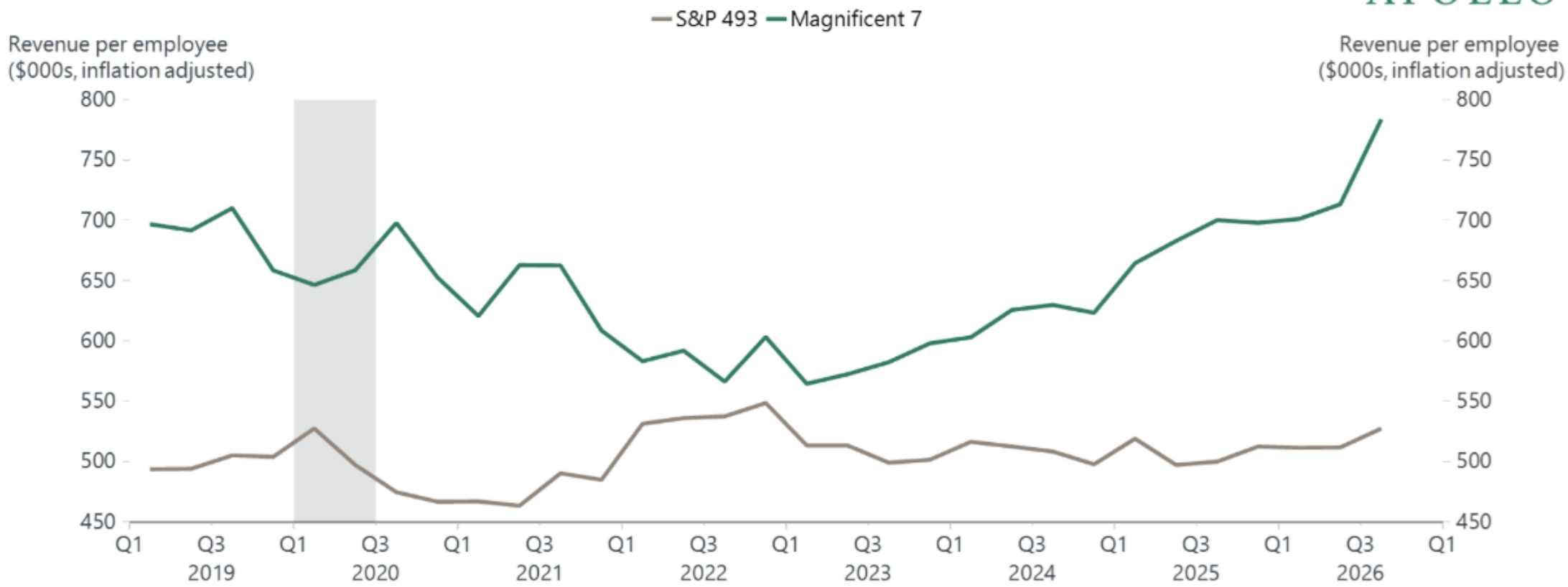
- Some results from research papers
  - AI assisted coding is 55 % faster on well defined coding task (GitHub controlled experiment, **2023**)
  - Experienced open-source developers worked with AI with project familiar to them. They belived they are faster with AI but actually they were 19 % slower (METR study, **Spring 2025**).
  - 84 % of developers use or plan to use AI tools. However, 46 % distrusts the accuracy of the output (Stack Overflow developer survey, 2025).

Productivity and AI

- Despite massive investments and hype around AI, the technology has delivered zero financial payoff to corporate America outside of tech.
- Revenue per worker is rising for the seven big technology companies (Mag 7) but is flat for the next 493 big companies in S&P 500 (S&P 493).
- **AI does not understand accountability, trust, culture, unwritten rules, ethical trade-offs or fairness.**
- **They shape reputational risk and ultimately drive revenue growth and profit margins.**
- Slok, T. (October 24, 2026). *No Signs of AI Payoff Outside the Tech Sector.* Apollo. https://www.apollo.com/wealth/insights-news/insights/daily-spark/no-signs-of-ai-payoff-outside-the-tech-sector

Revenue per worker moving up for tech but not for the rest of the economy

APOLLO



Discussion

- How would you measure productivity?
- Is work more productive if it produces more code?
- Should productivity be measured in euros?
- Should other factors be taken into account as well?

Tools

Agents in the editor (IDE)

GitHub Copilot agent mode, Cursor

Work with you inside VS Code or a similar editor

Terminal agents

Claude Code, OpenAI Codex CLI, Gemini CLI

Run in the command line and work on the whole project

Background / cloud agents

GitHub Copilot coding agent, OpenAI Codex, etc.

Take a ticket, work on their own and return a pull request

App builders for non-programmers

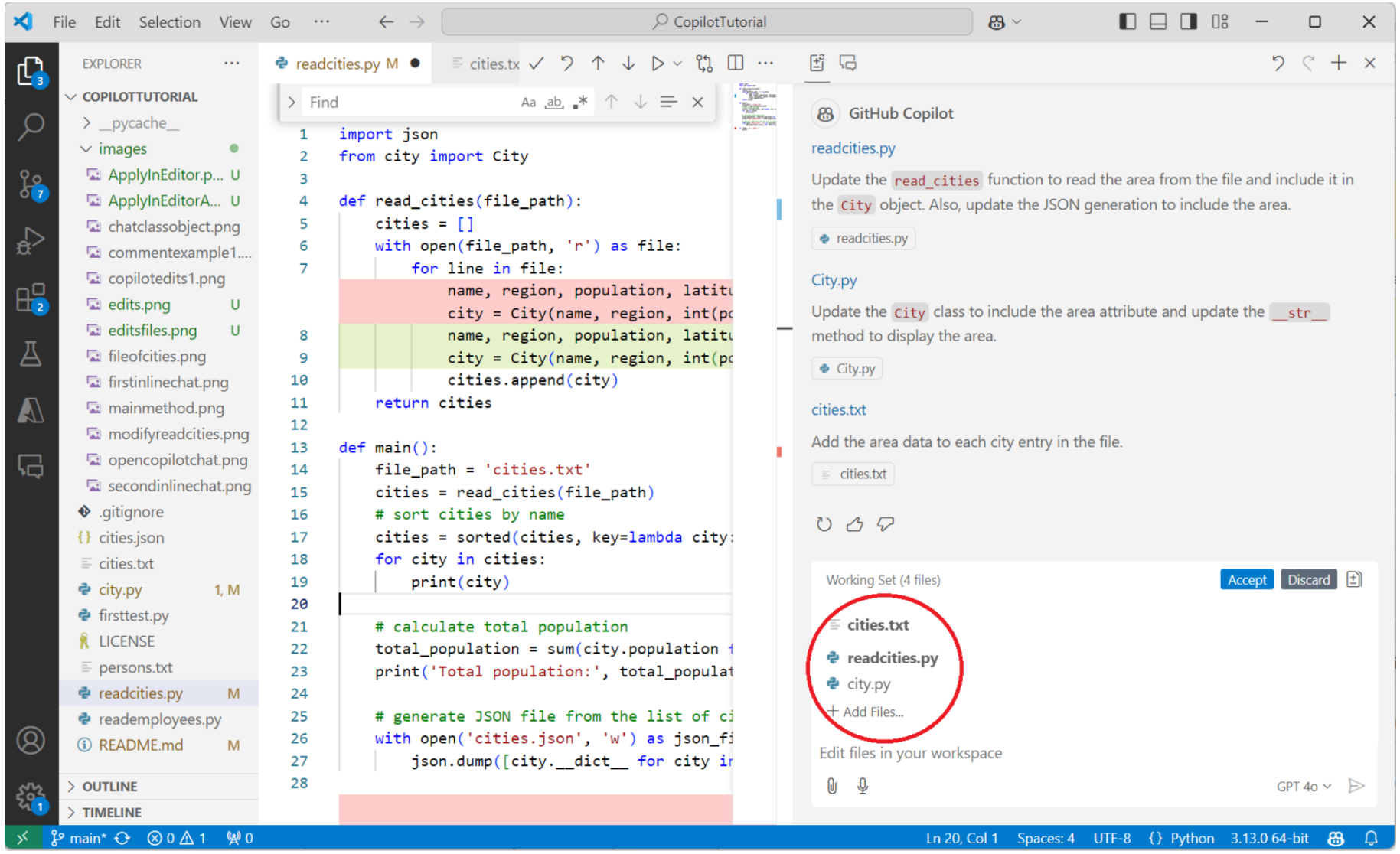
Lovable, etc.

Build and publish a web app from a description

Chat assistants ChatGPT, Claude, Microsoft Copilot, Gemini can also write and run small programs

Example: agent mode in VS Code

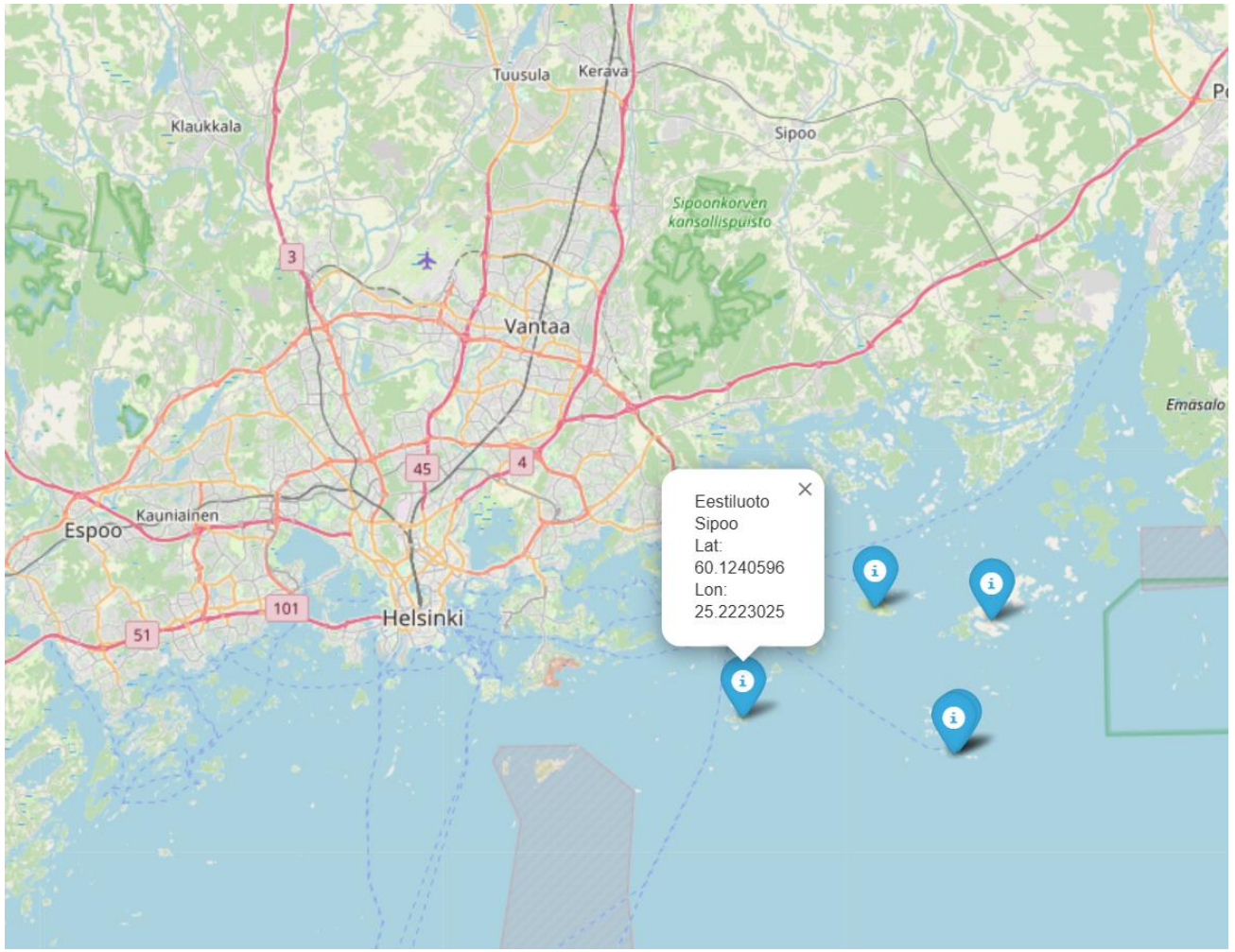
- In Agent mode you give a whole task to GitHub Copilot.
- The agent edits multiple files, runs terminal commands and fixes its own errors.
- You review the changes and keep or undo them.



Exercise 1

Exercise 1 – Map app with an agent

- We will build a program that shows interesting places on a map.
- In this exercise you do not write the code. Instead, you describe what you want and check the result.
- You can use any AI tool for programming mentioned earlier
- This is not agentic development with real tools, but it simulates the idea



Exercise 1 – Step 1

- Give the agent a specification
  - Open VS Code in Agent mode (or any other tool, or a chat assistant in the browser)
  - Give it the text below:

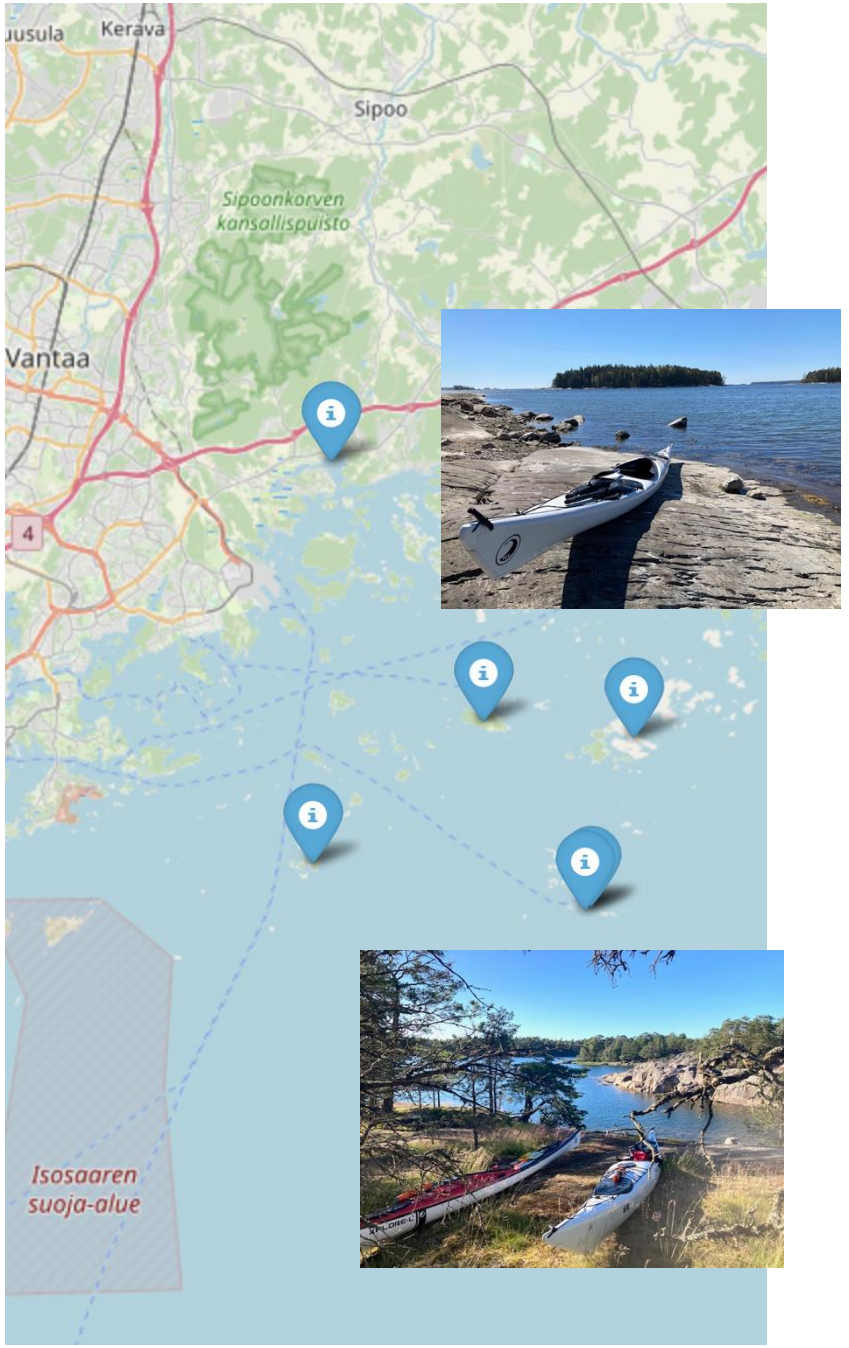
Goal: a Python program that shows recreational islands on a map.

Places (name, city, notes): Sipoonranta, Sipoo, Kayaking base; Kaunissaari, Sipoo, Beautiful islands with services; Pirttisaari, Porvoo, Old artillery base; Majakkasaari, Porvoo, Lighthouse island; Luotsisaari, Porvoo, Old pilot base; Eestiluoto, Sipoo, Nice islands far away

Requirements: find the coordinates with the OpenStreetMap Nominatim or similar service. Draw the islands on an interactive map (folium) saved as islands_map.html. Clicking a marker shows the name and notes.

Acceptance criteria: all six islands are in the sea near Sipoo and Porvoo. The program runs without errors.

First show me your plan. Then build and run the program.



Exercise 1 – Step 2: Build and run

- Read the plan generated by AI. Approve it or ask the “agent” to change it.
- Let the agent write and run the program
  - In VS Code the agent runs: python islands.py
  - Browser assistants can run the code themselves
- If something fails, do not fix the code yourself. Tell the agent what is wrong.
- Open islands_map.html in your browser.

Exercise 1 – Step 3: Review

- **Check:** Go through every acceptance criterion. Are all six islands in the right place?
- **Question:** Ask the agent: “Explain what your program does and what could go wrong.”
- **Change:** Add a feature by changing the specification, not the code – e.g. a different marker colour for islands with services.
- **Reflect:** What did the agent get wrong? How did you notice? Who is responsible for the result?

Advanced exercise for programmers

Build a web application with a coding agent

Claude Code, Codex, GitHub Copilot agent mode or Cursor

Backend Node.js + Express with the list of islands. Coordinates from Nominatim.

Frontend React + Leaflet showing the places on an OpenStreetMap map.

Style as in fullstackopen.com

Work agentically

- Write an AGENTS.md (or CLAUDE.md) first: project rules, how to run and test (let AI to generate this)
- Ask for a plan before any code
- Require automated tests
- (use Git)

Ethics of AI in software development

Exercise

- Consider ethical issues related to AI-assisted and agentic coding
- You may consider these issues, for example:
  - Accountability: who is responsible for code an agent wrote?
  - Environmental impact: an agent task uses far more computing than one chat answer
  - Security and privacy of the application and of the data the agent can access
  - Jobs and learning: how will young developers learn?
- You may read this:
  - https://merge.rocks/blog/the-ethics-of-ai-in-software-development-what-developers-need-to-know



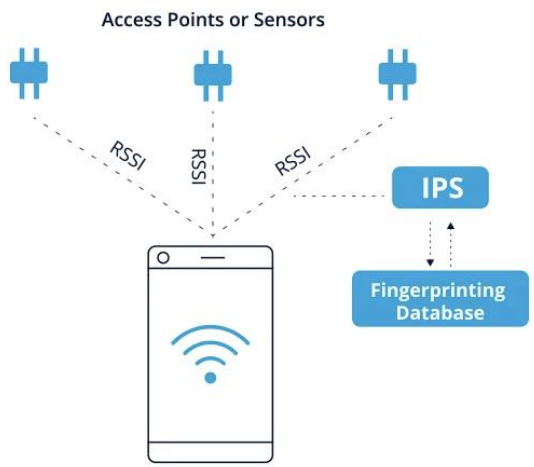
In 2026, Google’s subsidiary cleared forest in Muhos and Kajaani for data centers before environmental impact assessments were complete. Authorities halted work, and police are investigating suspected environmental damage at the Muhos site. 330 hectars of forest was destroyed (3 km x 1 km).

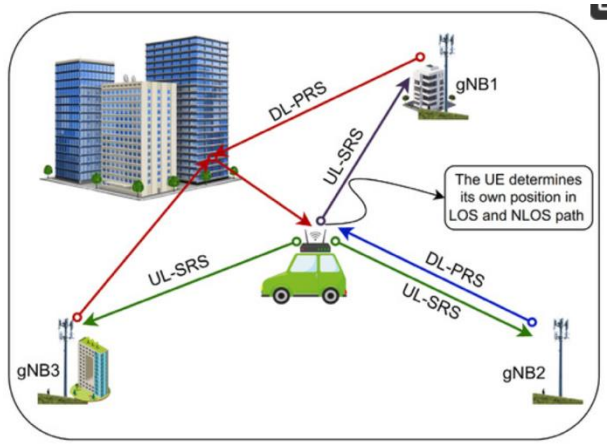
Location-based services

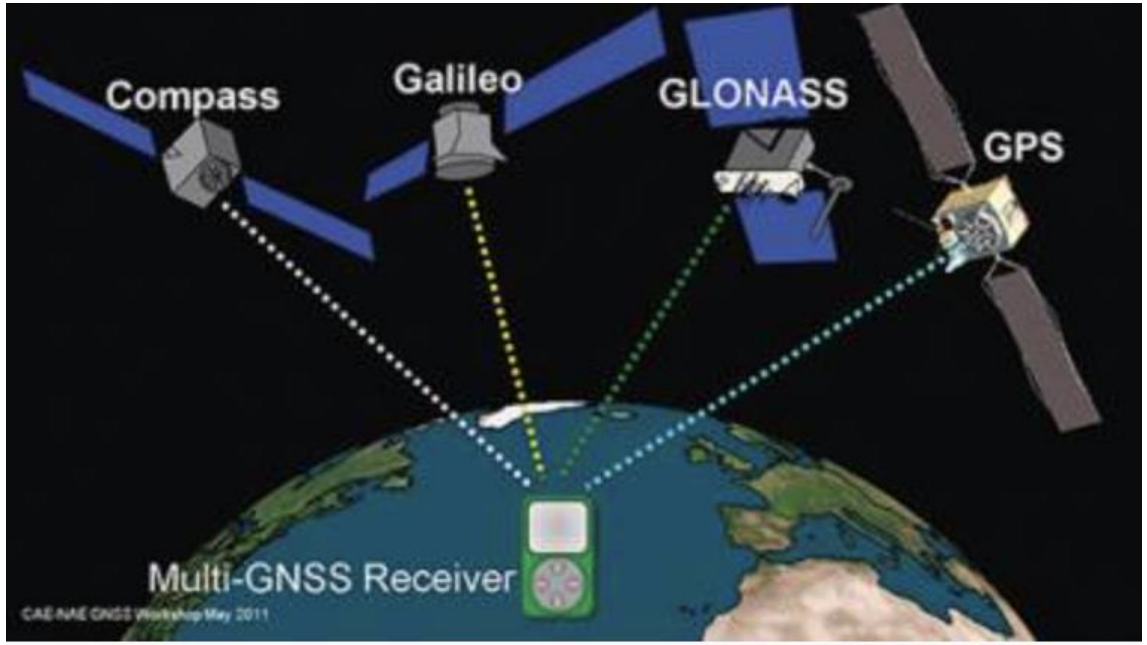
- Location-based services (LBS) are applications that use a user's geographic location to provide information and services.
- Examples
  - Navigation and maps: Apps like Google Maps provide directions and real-time traffic information
  - Local search: Services that help you find businesses or points of interest in your vicinity Social networking: friends who are nearby
  - Marketing: LBS to send targeted promotions to customers who are close to their store (real-time bidding (RTB))
  - Asset tracking: Track the location of their vehicles or equipment to manage their fleet more efficiently

Gathering and using location data

- The position of the user can be estimated with the following techniques
  - GNSS (=GPS, Galileo, Beidou, Glonass)
  - Cellular phone network (Cell ID)
  - Indoor positioning technologies, such as Wi-Fi, Bluetooth, UWB, etc.







Location data brokers

- Many location-based services collect the location data of the users. Sometimes, the intention is to sell the data to third parties, such as adver
- There are location data brokers, such as Gravy Analytics, which provide gathered location data to other companies for real-time bidding.
- The location data brokers gather the user position data from other applications such as SportTracker, MyFitnessPal and Tinder
- The location data is sold usually for marketing purposes. However, that data can be used also for to track people.

Ethical concerns of location-based services

- Privacy violations: LBS collect highly sensitive location data, which can be a privacy risk.
- Lack of transparency and control: Users may not understand how their data is being collected, used, or shared,
- Security risks: Location data can be vulnerable to breaches and unauthorized access.
- Algorithmic bias
- Surveillance and misuse: The data can be used for surveillance by governments or private companies
- Consumer mistrust: Lack of transparency and data misuse can damage a company's reputation

Location data of people can be purchased easily

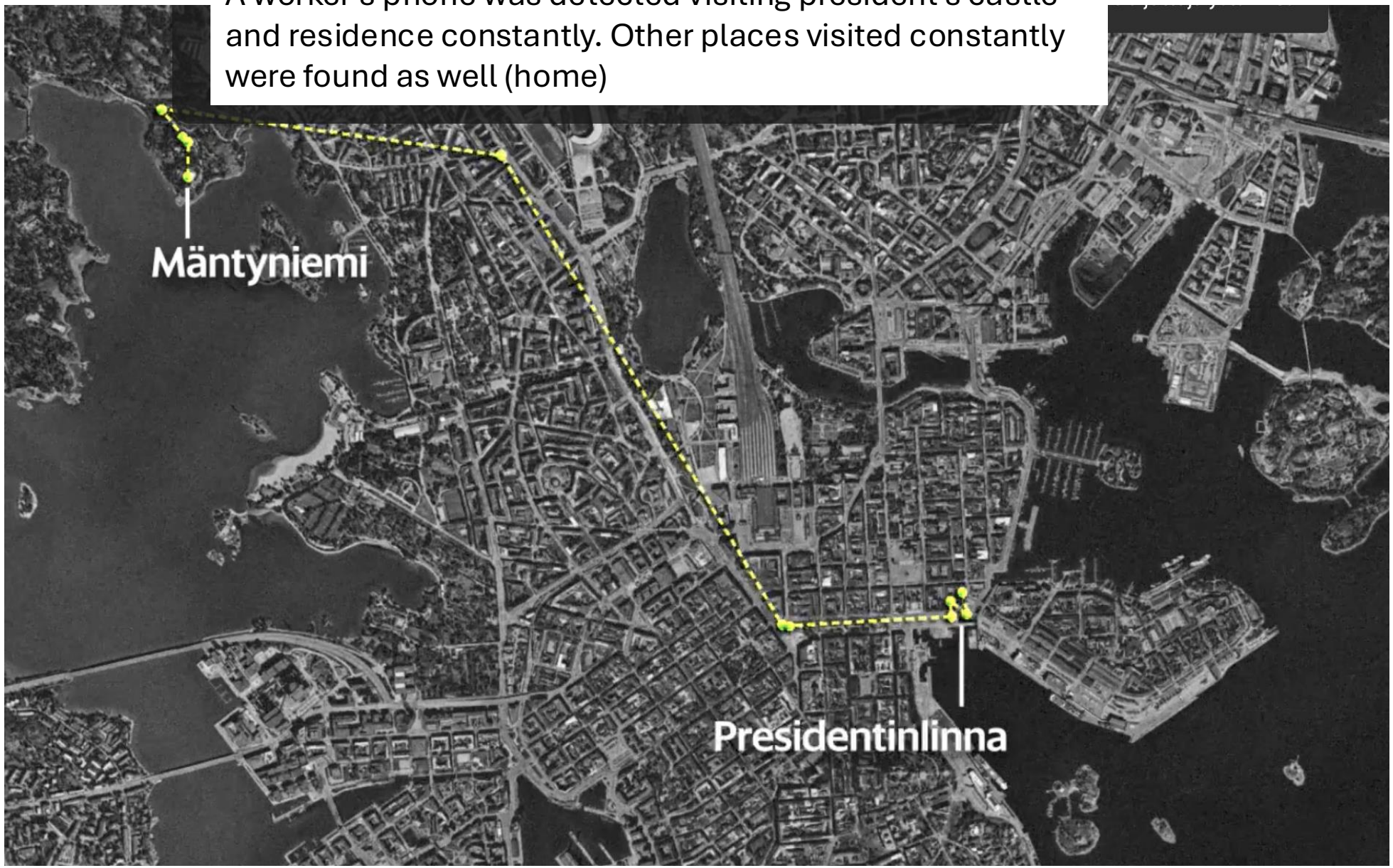
- Finnish newspaper Helsingin Sanomat obtained location data from a data broker covering nearly one million Finnish phones.
- Journalists were able to identify persons from anonymized data by analyzing the location data of people. Politicians, military personnel, people close to president, etc.
- The data shows the movement of persons travelling between working places, homes, schools, hobby places, etc.
- A European company offered continuously updated location data from over 1,8 million Finnish phones for 42 000 € a year. The data is available also from all the Europe.

https://www.hs.fi/tutkiva/art-2000011978813.html

Location data of people is for sale



A worker’s phone was detected visiting president’s castle and residence constantly. Other places visited constantly were found as well (home)



Discussion

- Consider ethical issues related to location-based services(LBS) (location data of people)
- Risk of de-anonymization and misuse for profiling. Combining location data with other sources
- Replacing real identifiers by pseudonyms is not enough to protect one’s privacy
- https://arxiv.org/pdf/2403.05558

Exercise

- Gravy Analytics, a location data broker, was hacked at January 2025.
- Significant data breach potentially exposed through de-anonymization the precise location information of millions of individuals.

Claude terminal output

CLAUDE.md

CLAUDE ⟩ Islands map – project rules

```
# Islands map – project rules

Small teaching demo for a BIP workshop on agentic software development.
The programs read location data and show it on interactive maps.
The audience includes business students, so the code must be easy to read.

## Tech stack
- Python 3, one main script: `islands.py`
- Dependencies listed in `requirements.txt` and installed into a local virtual environment `.venv`
- Maps with `folium`, output as a static HTML file

## How to run
```bash
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt
.venv/bin/python islands.py
```

Coding rules
- Keep it simple: small functions, clear names, short comments in English.
- No hidden magic: print what the program is doing (e.g. "Looking up Kaunissaari, Sipoo ... found").
- Never install packages globally and never touch files outside this folder.

Geocoding rules (OpenStreetMap Nominatim)
- Follow the Nominatim usage policy: a descriptive User-Agent and at most 1 request per second.
- Cache results in `coordinates.json` so that reruns do not call the service again.
- If a place is not found, or the result is clearly in the wrong place, say so in the output.
  Do not silently hard-code coordinates. A fallback is allowed only if it is reported.

How to verify your work
- Run the program after every change and read its output.
- Sanity check: every place must be in the Sipoo–Porvoo coastal area
  (roughly latitude 60.05–60.45, longitude 25.05–25.90). Report any place outside it.
- Confirm that `islands_map.html` was created and has one marker per place.

Way of working
- Show a short plan before writing code and wait for approval.
- Work in small steps and summarise at the end what was done and what was not verified.
```

```
❯ /login
  └ Login successful

● You're already logged in — the login succeeded. You're good to
  continue working; just let me know what you'd like to do next in the
  islands-agent project.

✻ Churned for 3s · done 10:18 PM

❯ cd ~/Documents/ClaudeBIP/islands-agent
  └ 3 skills available

  Ran 1 shell command

● Now in ~/Documents/ClaudeBIP/islands-agent. It currently only has
  CLAUDE.md — no islands.py or requirements.txt yet. Want me to set up
  the project (per the CLAUDE.md rules: islands.py, requirements.txt,
  venv) with a short plan first?

✻ Worked for 6s · done 10:19 PM

❯ Goal: a Python program that shows recreational islands on a map.
  Places (name, city, notes): Sipoonranta, Sipoo, Kayaking base;
  Kaunissaari, Sipoo, Beautiful islands with services; Pirttisaari,
  Porvoo, Old artillery base; Majakkasaari, Porvoo, Lighthouse island;
  Luotsisaari, Porvoo, Old pilot base; Eestiluoto, Sipoo, Nice islands
  far away
  Requirements: find the coordinates with the OpenStreetMap Nominatim
  or similar service. Draw the islands on an interactive map (folium)
  saved as islands_map.html. Clicking a marker shows the name and
  notes.
  Acceptance criteria: all six islands are in the sea near Sipoo and
  Porvoo. The program runs without errors.
  First show me your plan. Then build and run the program.
```

● **Plan**

1. Project setup
- requirements.txt: folium, requests
- Create .venv, install deps

2. islands.py structure
- List of 6 places as dicts: name, city, notes
- geocode(place, city) — query Nominatim (https://nominatim.openstreetmap.org/search), params q=f"{place}, {city}, Finland", format=json, limit=1; descriptive User-Agent header; sleep 1s between requests
- Cache: load/save coordinates.json keyed by "name, city", so reruns skip Nominatim entirely
- Print progress per place: "Looking up Kaunissaari, Sipoo ... found (60.xx, 25.xx)" or "... NOT FOUND"
- Sanity check each result against the bounding box (lat 60.05–60.45, lon 25.05–25.90); print a warning if outside, but still use it (reported, not hidden)
- Build a folium.Map centered on the average coordinate, add one Marker per place with a popup showing name + notes (HTML), save as islands_map.html

3. Run & verify
- Run the script, read console output for each lookup
- Confirm all 6 are within the bounding box (flag any that aren't)
- Confirm islands_map.html exists with 6 markers

Shall I go ahead and build this?

✻ Cogitated for 8s · done 10:20 PM

❯ yes

● **Write**(requirements.txt)
  ⎿ Wrote **2** lines to **requirements.txt**
```
1 folium
2 requests
```

● **Write**(islands.py)
  ⎿ Wrote **142** lines to **islands.py**
```
1 """Show recreational islands near Sipoo and Porvoo on an
  interactive map.
2
  vice, caches them
4 in coordinates.json, and draws the islands with folium ma
  rkers.
5 """
7 import json
8 import os
9 import time
… +159 lines
```

● Now setting up the venv, installing dependencies, and running the script.

  Ran **2** shell commands

● Done. Summary:

  - Built islands.py, requirements.txt, .venv as planned.
  - Ran successfully: all 6 islands geocoded via Nominatim (cached in coordinates.json), all within the Sipoo–Porvoo bounding box — no warnings.
  - islands_map.html created with 6 markers, each with a popup showing name + notes.

  Not verified: I haven't opened the HTML in a browser to visually confirm the popups render correctly on click — only confirmed 6 marker entries exist in the file. Want me to open it to check visually?

✻ Cooked for 1m 41s · done 10:22 PM

❯ yes